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Module 3 Introduction
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Graded Assignment: Learning Check - Search Problem Formulation
Submitted
Lesson 3: Advanced Search Algorithms
Module Assignment

Your grade: 100%

Your latest: 100% • Your highest: 100% • To pass you need at least 80%. We keep your highest score.

Next item

Learning Check - Search Problem Formulation

Instructions

Review Learning Objectives

1. Which of the following are necessary to formally define a problem for a problem-solving agent? Select all that apply. **1 / 1 point**

coach
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- [x] Specifying the outcome of each action that the agent could take

Correct
A problem is defined by: states, initial state, actions, transition model, path cost, goal test.

- [x] Whether the agent has reached its goal

Correct
A problem is defined by: states, initial state, actions, transition model, path cost, goal test.

- [ ] The list of all goals that the agent could potentially attempt
- [x] The initial state the agent starts in

Correct
A problem is defined by: states, initial state, actions, transition model, path cost, goal test.

Assignment details
Due: Oct 4, 11:59 PM IST
Attempts: 3 left (5 attempts every 24 hours)
Try again
Submitted: Sep 29, 1:44 AM IST
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2. Which of the following statements about tree search and graph search are correct? Select all that apply. **1 / 1 point**

- [x] If a state is reachable via multiple paths, tree search will create two separate nodes corresponding to ea ch path.

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Correct
Tree search treats different paths to the same node as distinct. Hence it runs into the problem of failure to detect repeated states, which can turn a linear problem into an exponential one.

- [ ] Tree search checks if a node has been visited before adding it to the search queue.
- [ ] Graph search is the same as tree search, just a different name of the same search algorithm.
- [x] Graph search checks if a node has been visited before adding it to the search queue.

Correct
This is the simple but important difference between the tree and graph search.

3. What is the major disadvantage of graph search as opposed to tree search? **1 / 1 point**

- ( ) Graph search is unable to detect repeated states
- (•) Graph search can require a lot of memory because it keeps track of all possible states
- ( ) Visual representations of a graph search are less appealing but graph search is otherwise the same as tr ee search.
- ( ) There is no disadvantage since graph search is just another name for tree search.

Correct
This is a drawback to maintaining the secondary data structure to monitor visited nodes